

Literature Review of RipVIS (Rip Currents Video Instance Segmentation Benchmark) - <https://ripvis.ai>

The authors introduce the RipVIS (Rip Currents Video Instance Segmentation) benchmark specifically to address the task of rip current segmentation. They provide pixel-level, polygonal annotations to capture the precise shapes of rip currents across video frames. The paper explicitly contrasts this task with bounding box detection, noting that while bounding boxes have been used in previous datasets, they lack the granularity needed for rip currents. Because,

- Rip currents are amorphous and constantly changing in shape.
- Bounding boxes often include too much background or leave out significant parts of the current.

Size / splits

- Total: 184 videos, 212,328 frames
- With rip currents: 150 videos, 163,528 frames
- Without rip currents (negative videos): 34 videos, 48,800 frames
- Train/Val/Test split (they explicitly include all three):
 - Train: 112 videos, 147,802 frames
 - Val: 36 videos, 32,566 frames
 - Test: 36 videos, 31,960 frames

Annotation type + label definitions

In this context, an annotation is the ground-truth label that humans add to an image or video frame so a model can learn and be evaluated.

What's annotated

- The benchmark is for **video instance segmentation** (mask overlay per instance).
- **Pixel-level annotations:**
 - **15,784 frames** manually annotated using **polygons**

- **25,298 rip-current instances** total
- **~1.6 rip currents per frame** average
- They generate **interpolated annotations** for intermediate frames (between manual ones) and verify them, so the dataset provides masks across the larger set of frames.
- The paper also states that (like the other datasets they compare to) **RipVIS includes bounding box annotations** too.

In RipVis, an annotation is typically a **polygon/mask** that traces the rip current region in a frame, turning it into a pixel-level label that the model can learn to reproduce.

Models and techniques used for detection

Category	Specific Models / Techniques	Description / Key Details
Two-Stage Detectors	Mask R-CNN Cascade Mask R-CNN	Baseline models that combine region proposals with pixel-level segmentation masks. Cascade Mask R-CNN extends this with a multi-stage refinement approach to progressively improve mask quality.
One-Stage Detectors	YOLO11 (nano, small, medium, large, extra-large) SparseInst(ResNet-50, ResNet-101, PVTv2-B1 backbones)	Fully convolutional models designed for real-time performance. YOLO11 incorporates a Cross-Stage Partial Spatial Attention (C2PSA) module for spatial sensitivity. SparseInst leverages sparse instance activation maps to segment objects directly without region proposals.
Core Post-Processing Technique	Temporal Confidence Aggregation (TCA)	A novel pixel-level post-processing method designed to stabilize frame-by-frame segmentation of amorphous, dynamic rip currents by aggregating confidence scores over time.
TCA Pipeline Components	Downsampling Instance Tracking Temporal Smoothing Hysteresis Thresholding	Masks are downsampled for efficiency. Instances are tracked across frames using Intersection over Union (IoU) and the Hungarian algorithm. A heatmap is maintained for each tracked instance

Explainability Approaches

Temporal HeatmapsMask Overlays

where scores increment with detection and decay in absence. Finally, a dual-threshold filtering technique determines the final object boundaries.

The TCA process actively computes a temporal heatmap that accumulates confidence scores. These heatmaps and the resulting refined segmentation masks are overlaid directly onto the video frames to visually explain the model's confidence and localization over time.

Special Components

The key special component introduced in the paper is a novel post-processing step called Temporal Confidence Aggregation (TCA). Because rip currents constantly change shape and intensity, frame-by-frame segmentation can be noisy.

What goes wrong without TCA

Even if a rip current is genuinely present for many seconds, a frame-by-frame model might:

- detect it strongly in one frame,
- miss it in the next (due to foam/noise/glare),
- detect it again after that,
- or the mask “jumps” around and changes shape a lot.

This can cause false alarms or create false negatives, which will reduce trust in the system.

TCA stabilizes detection by aggregating confidence scores across video frames. The TCA pipeline includes:

- Downsampling(making something **smaller / lower resolution** than the original.): Prediction masks are downsampled to reduce computational complexity while preserving spatial relationships.
- Instance Tracking: The system tracks instances across consecutive frames using Intersection over Union (IoU) and the Hungarian algorithm for consistent identity assignment.

- **Temporal Smoothing:** A heatmap is maintained for each tracked instance. Pixel confidence scores incrementally increase with repeated detections across frames and gradually decay if the rip current is no longer detected.
- **Hysteresis Thresholding:** To output the final mask, the method applies a dual-threshold filtering technique (adapted from the Canny edge detector) to the accumulated heatmaps to decide which pixels form the final object

Explainability Approach

The study uses **temporal heatmaps and mask overlays** for visualization and explainability.

- **Heatmaps:** The TCA process actively computes a temporal heatmap that accumulates confidence scores over time. The authors provide visual figures showing these computed heatmaps overlaid directly onto the video frames to intuitively represent where the model is confident a rip current is located.
- **Overlays:** The paper also displays the original model prediction masks and the TCA-refined prediction masks overlaid on the original beach footage, side-by-side with ground truth polygons, to visualize the smoothing and false-negative recovery enabled by their method

Metrics Reported

The authors utilize several standard video instance segmentation metrics to evaluate their models, including:

- Precision and Recall.
- Average Precision (AP) and AP50, which assess detection quality across varying confidence thresholds. The authors note that because their model only detects a single class (rip currents), the Mean Average Precision (mAP) is equivalent to the AP.
- Intersection over Union (IoU), used to evaluate spatial accuracy.
- F-beta scores (F1 and F2). The study places a particular emphasis on the F2 score because it heavily prioritizes recall. In the safety-critical context of beach monitoring, minimizing false negatives (missed rip currents) is considered far more important than minimizing false positives.
- Frames Per Second (FPS), to evaluate the real-time processing speed of the models.

Baseline comparisons

To establish baseline comparisons, the authors evaluate several popular state-of-the-art segmentation models, categorizing them into two architectures:

- Two-Stage Detectors: Mask R-CNN and Cascade Mask R-CNN.
- One-Stage Detectors: Five variants of YOLO11 (nano, small, medium, large, and extra-large) and two variants of SparseInst (utilizing ResNet-50 and transformer-based PVTv2 backbones).

In their experiments, they run these baseline models both in their original state and combined with Temporal Confidence Aggregation (TCA). This side-by-side comparison is used to demonstrate how their novel post-processing technique reduces noise, recovers false negatives, and significantly improves F2 scores across all baseline models.

Headline Results

- Best Overall Performance: The SparseInst PVTv2 model combined with Temporal Confidence Aggregation (TCA) achieved the best safety-critical metrics, yielding an F2 score of 0.780 and a Recall of 0.798.
- Highest Precision: The YOLO11l (large) variant with TCA achieved the highest Precision at 0.819.
- Fastest Speed: For real-time processing, the original YOLO11n (nano) model without TCA was the fastest, running at 128.20 FPS.

Failure Cases The amorphous and dynamic nature of rip currents leads to several challenging detection scenarios:

- Environmental & Lighting Factors: The authors note that rip currents can be temporarily obscured or rendered less visible due to lighting or environmental factors, making them impossible to detect even for expert annotators in certain frames.
- Turbulent Water: Intermittent appearances of rip currents in challenging, turbulent areas of water make consistent segmentation highly difficult.
- Camera Movement Issues: A specific failure case introduced by the TCA post-processing technique occurs during fast camera movement. If the camera suddenly pans after a stationary detection, the temporal smoothing can lag, which reduces detection accuracy and increases the number of false positives.